DBMS LAB ASSIGNMENT-3

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Software: MySQL Workbench
Database: DBMS_LAB_ASS3
Note: The assignment uses Oracle-style data types. The queries below use MySQL-compatible types and syntax. Run the questions in order in the same database.

QUESTION 1: Create the EMPLOYEE table

Create the database, select it, and create the EMPLOYEE table with the fields specified in the assignment.

SQL Query:

```
CREATE DATABASE DBMS_LAB_ASS3;
USE DBMS_LAB_ASS3;

CREATE TABLE EMPLOYEE (
    EMPLOYEE_ID INT,
    LAST_NAME VARCHAR(25),
    JOB_ID VARCHAR(10),
    SALARY DECIMAL(8,2),
    COMM_PCT DECIMAL(4,2),
    MGR_ID INT,
    DEPARTMENT_ID INT
);

DESC EMPLOYEE;
```

Output:

```
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| EMPLOYEE_ID   | int          | YES  |     | NULL    |       |
| LAST_NAME     | varchar(25)  | YES  |     | NULL    |       |
| JOB_ID        | varchar(10)  | YES  |     | NULL    |       |
| SALARY        | decimal(8,2) | YES  |     | NULL    |       |
| COMM_PCT      | decimal(4,2) | YES  |     | NULL    |       |
| MGR_ID        | int          | YES  |     | NULL    |       |
| DEPARTMENT_ID | int          | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
```

The CREATE DATABASE and CREATE TABLE statements complete successfully. DESC displays the table structure; no employee records have been inserted yet.

QUESTION 2: Insert the given employee records

Insert all 11 records supplied in the assignment. Use NULL for missing commission percentages.

SQL Query:

```
INSERT INTO EMPLOYEE
(EMPLOYEE_ID, LAST_NAME, JOB_ID, SALARY, COMM_PCT, MGR_ID, DEPARTMENT_ID)
VALUES
(198, 'Connell', 'SH_CLERK', 2600, 2.5, 124, 50),
(199, 'Grant', 'SH_CLERK', 2600, 2.2, 124, 50),
(200, 'Whalen', 'AD_ASST', 4400, 1.3, 101, 10),
(201, 'Hartstein', 'IT_PROG', 6000, NULL, 100, 20),
(202, 'Fay', 'AC_MGR', 6500, NULL, 210, 20),
(203, 'Mavris', 'AD_VP', 7500, NULL, 101, 40),
(204, 'Baer', 'AD_PRES', 3500, 1.5, 101, 90),
(205, 'Higgins', 'AC_MGR', 2300, NULL, 101, 60),
(206, 'Gitz', 'IT_PROG', 5000, NULL, 103, 60),
(100, 'King', 'AD_ASST', 8956, 0.3, 108, 100),
(101, 'Kochar', 'SH_CLERK', 3400, 1.3, 118, 30);

SELECT * FROM EMPLOYEE;
```

Output:

```
Query OK, 11 rows affected

+-------------+-----------+----------+---------+----------+--------+---------------+
```

```
| EMPLOYEE_ID | LAST_NAME | JOB_ID   | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
+-------------+-----------+----------+---------+----------+--------+---------------+
| 198         | Connell   | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
| 199         | Grant     | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
| 200         | Whalen    | AD_ASST  | 4400.00 | 1.30     | 101    | 10            |
| 201         | Hartstein | IT_PROG  | 6000.00 | NULL     | 100    | 20            |
| 202         | Fay       | AC_MGR   | 6500.00 | NULL     | 210    | 20            |
| 203         | Mavris    | AD_VP    | 7500.00 | NULL     | 101    | 40            |
| 204         | Baer      | AD_PRES  | 3500.00 | 1.50     | 101    | 90            |
| 205         | Higgins   | AC_MGR   | 2300.00 | NULL     | 101    | 60            |
| 206         | Gitz      | IT_PROG  | 5000.00 | NULL     | 103    | 60            |
| 100         | King      | AD_ASST  | 8956.00 | 0.30     | 108    | 100           |
| 101         | Kochar    | SH_CLERK | 3400.00 | 1.30     | 118    | 30            |
+-------------+-----------+----------+---------+----------+--------+---------------+
```

The SELECT output is shown in the same order as the inserted records.

QUESTION 3: Display employee ID, last name and job ID

Display the three requested columns, with EMPLOYEE_ID appearing first.

SQL Query:

```
SELECT EMPLOYEE_ID, LAST_NAME, JOB_ID
FROM EMPLOYEE;
```

Output:

```
+-------------+-----------+----------+
| EMPLOYEE_ID | LAST_NAME | JOB_ID   |
+-------------+-----------+----------+
| 198         | Connell   | SH_CLERK |
| 199         | Grant     | SH_CLERK |
| 200         | Whalen    | AD_ASST  |
| 201         | Hartstein | IT_PROG  |
| 202         | Fay       | AC_MGR   |
| 203         | Mavris    | AD_VP    |
| 204         | Baer      | AD_PRES  |
| 205         | Higgins   | AC_MGR   |
| 206         | Gitz      | IT_PROG  |
| 100         | King      | AD_ASST  |
| 101         | Kochar    | SH_CLERK |
+-------------+-----------+----------+
```

QUESTION 4: Display all employees of department 60

Use WHERE to select employees whose DEPARTMENT_ID is 60.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE DEPARTMENT_ID = 60;
```

Output:

```
+-------------+----------+---------+---------+----------+--------+---------------+
| EMPLOYEE_ID | LAST_NAME| JOB_ID  | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
+-------------+----------+---------+---------+----------+--------+---------------+
| 205         | Higgins  | AC_MGR  | 2300.00 | NULL     | 101    | 60            |
| 206         | Gitz     | IT_PROG | 5000.00 | NULL     | 103    | 60            |
+-------------+----------+---------+---------+----------+--------+---------------+
```

QUESTION 5: Display details of the employee whose last name is King

Select the employee record where LAST_NAME is 'King'.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE LAST_NAME = 'King';
```

Output:

```
+-------------+-----------+---------+---------+----------+--------+---------------+
 | EMPLOYEE_ID | LAST_NAME | JOB_ID  | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
 +-------------+-----------+---------+---------+----------+--------+---------------+
 | 100         | King      | AD_ASST | 8956.00 | 0.30     | 108    | 100           |
 +-------------+-----------+---------+---------+----------+--------+---------------+
```

QUESTION 6: Display unique JOB_ID values as JOB_TITLE

DISTINCT removes duplicate job IDs. AS gives the output column the alias JOB_TITLE.

SQL Query:

```
SELECT DISTINCT JOB_ID AS JOB_TITLE
FROM EMPLOYEE;
```

Output:

```
+-----------+
 | JOB_TITLE |
 +-----------+
 | SH_CLERK  |
 | AD_ASST   |
 | IT_PROG   |
 | AC_MGR    |
 | AD_VP     |
 | AD_PRES   |
 +-----------+
```

QUESTION 7: Display salary with an increase of Rs. 300

Display LAST_NAME, SALARY and the calculated salary after adding 300. The alias is Increased Salary.

SQL Query:

```
SELECT LAST_NAME, SALARY,
       SALARY + 300 AS `Increased Salary`
FROM EMPLOYEE;
```

Output:

```
+-----------+---------+------------------+
| LAST_NAME | SALARY  | Increased Salary |
+-----------+---------+------------------+
| Connell   | 2600.00 | 2900.00          |
| Grant     | 2600.00 | 2900.00          |
| Whalen    | 4400.00 | 4700.00          |
| Hartstein | 6000.00 | 6300.00          |
| Fay       | 6500.00 | 6800.00          |
| Mavris    | 7500.00 | 7800.00          |
| Baer      | 3500.00 | 3800.00          |
| Higgins   | 2300.00 | 2600.00          |
| Gitz      | 5000.00 | 5300.00          |
| King      | 8956.00 | 9256.00          |
| Kochar    | 3400.00 | 3700.00          |
+-----------+---------+------------------+
```

QUESTION 8: Display annual compensation with a one-time bonus of Rs. 100

Annual compensation is calculated as monthly salary multiplied by 12, plus a one-time bonus of 100.

SQL Query:

```
SELECT LAST_NAME, SALARY,
       (SALARY * 12) + 100 AS `Annual Compensation`
FROM EMPLOYEE;
```

Output:

```
+-----------+---------+--------------------+
| LAST_NAME | SALARY  | Annual Compensation |
+-----------+---------+--------------------+
| Connell   | 2600.00 | 31300.00           |
| Grant     | 2600.00 | 31300.00           |
| Whalen    | 4400.00 | 52900.00           |
| Hartstein | 6000.00 | 72100.00           |
| Fay       | 6500.00 | 78100.00           |
| Mavris    | 7500.00 | 90100.00           |
| Baer      | 3500.00 | 42100.00           |
| Higgins   | 2300.00 | 27700.00           |
| Gitz      | 5000.00 | 60100.00           |
| King      | 8956.00 | 107572.00          |
| Kochar    | 3400.00 | 40900.00           |
+-----------+---------+--------------------+
```

QUESTION 9: Display employees who get commission

Employees who get commission have a COMM_PCT value that is not NULL.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE COMM_PCT IS NOT NULL;
```

Output:

```
+-------------+---------+----------+---------+----------+--------+---------------+
| EMPLOYEE_ID | LAST_NAME| JOB_ID  | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
+-------------+---------+----------+---------+----------+--------+---------------+
| 198         | Connell | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
```

```
| 199         | Grant   | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
| 200         | Whalen  | AD_ASST  | 4400.00 | 1.30     | 101    | 10            |
| 204         | Baer    | AD_PRES  | 3500.00 | 1.50     | 101    | 90            |
| 100         | King    | AD_ASST  | 8956.00 | 0.30     | 108    | 100           |
| 101         | Kochar  | SH_CLERK | 3400.00 | 1.30     | 118    | 30            |
+-------------+---------+----------+---------+----------+--------+---------------+
```

QUESTION 10: Display employees who do not get commission

Employees who do not get commission have COMM_PCT set to NULL.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE COMM_PCT IS NULL;
```

Output:

```
+-------------+-----------+---------+---------+----------+--------+---------------+
 | EMPLOYEE_ID | LAST_NAME | JOB_ID  | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
 +-------------+-----------+---------+---------+----------+--------+---------------+
 | 201         | Hartstein | IT_PROG | 6000.00 | NULL     | 100    | 20            |
 | 202         | Fay       | AC_MGR  | 6500.00 | NULL     | 210    | 20            |
 | 203         | Mavris    | AD_VP   | 7500.00 | NULL     | 101    | 40            |
 | 205         | Higgins   | AC_MGR  | 2300.00 | NULL     | 101    | 60            |
 | 206         | Gitz      | IT_PROG | 5000.00 | NULL     | 103    | 60            |
 +-------------+-----------+---------+---------+----------+--------+---------------+
```

QUESTION 11: Display employee ID, department ID and salary above 5000

Select only the requested columns for employees whose salary is greater than 5000.

SQL Query:

```
SELECT EMPLOYEE_ID, DEPARTMENT_ID, SALARY
FROM EMPLOYEE
WHERE SALARY > 5000;
```

Output:

```
+-------------+---------------+---------+
| EMPLOYEE_ID | DEPARTMENT_ID | SALARY  |
+-------------+---------------+---------+
| 201         | 20            | 6000.00 |
| 202         | 20            | 6500.00 |
| 203         | 40            | 7500.00 |
| 100         | 100           | 8956.00 |
+-------------+---------------+---------+
```

QUESTION 12: Display last name and salary between 4000 and 7000

BETWEEN includes both boundary values, 4000 and 7000.

SQL Query:

```
SELECT LAST_NAME, SALARY
FROM EMPLOYEE
WHERE SALARY BETWEEN 4000 AND 7000;
```

Output:

```
+-----------+---------+
| LAST_NAME | SALARY  |
+-----------+---------+
| Whalen    | 4400.00 |
| Hartstein | 6000.00 |
| Fay       | 6500.00 |
+-----------+---------+
```

QUESTION 13: Display employees whose salary is 6000, 6500 or 7000

IN checks whether SALARY matches any value in the given list. There is no employee with salary 7000 in the supplied data.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE SALARY IN (6000, 6500, 7000);
```

Output:

```
+-------------+-----------+---------+---------+----------+--------+---------------+
 | EMPLOYEE_ID | LAST_NAME | JOB_ID  | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
 +-------------+-----------+---------+---------+----------+--------+---------------+
```

```
| 201         | Hartstein | IT_PROG | 6000.00 | NULL     | 100    | 20            |
| 202         | Fay       | AC_MGR  | 6500.00 | NULL     | 210    | 20            |
+-------------+-----------+---------+---------+----------+--------+---------------+
```

QUESTION 14: Display employees working in departments 10, 20, 30 or 50

IN selects employees whose DEPARTMENT_ID matches one of the listed department numbers.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE DEPARTMENT_ID IN (10, 20, 30, 50);
```

Output:

```
+-------------+-----------+----------+---------+----------+--------+---------------+
| EMPLOYEE_ID | LAST_NAME | JOB_ID   | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
+-------------+-----------+----------+---------+----------+--------+---------------+
| 198         | Connell   | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
| 199         | Grant     | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
| 200         | Whalen    | AD_ASST  | 4400.00 | 1.30     | 101    | 10            |
| 201         | Hartstein | IT_PROG  | 6000.00 | NULL     | 100    | 20            |
| 202         | Fay       | AC_MGR   | 6500.00 | NULL     | 210    | 20            |
| 101         | Kochar    | SH_CLERK | 3400.00 | 1.30     | 118    | 30            |
+-------------+-----------+----------+---------+----------+--------+---------------+
```

QUESTION 15: Display employees whose salary is not equal to 5000

The <> operator means 'not equal to'. The employee with salary 5000 is excluded.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE SALARY <> 5000;
```

Output:

```
+-------------+-----------+----------+---------+----------+--------+---------------+
| EMPLOYEE_ID | LAST_NAME | JOB_ID   | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
+-------------+-----------+----------+---------+----------+--------+---------------+
| 198         | Connell   | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
| 199         | Grant     | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
| 200         | Whalen    | AD_ASST  | 4400.00 | 1.30     | 101    | 10            |
| 201         | Hartstein | IT_PROG  | 6000.00 | NULL     | 100    | 20            |
| 202         | Fay       | AC_MGR   | 6500.00 | NULL     | 210    | 20            |
| 203         | Mavris    | AD_VP    | 7500.00 | NULL     | 101    | 40            |
| 204         | Baer      | AD_PRES  | 3500.00 | 1.50     | 101    | 90            |
| 205         | Higgins   | AC_MGR   | 2300.00 | NULL     | 101    | 60            |
| 100         | King      | AD_ASST  | 8956.00 | 0.30     | 108    | 100           |
| 101         | Kochar    | SH_CLERK | 3400.00 | 1.30     | 118    | 30            |
+-------------+-----------+----------+---------+----------+--------+---------------+
```

QUESTION 16: Display details of all clerks

In the supplied data, SH_CLERK is the job ID used for clerks.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE JOB_ID = 'SH_CLERK';
```

Output:

```
+-------------+---------+----------+---------+----------+--------+---------------+
| EMPLOYEE_ID | LAST_NAME| JOB_ID   | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
+-------------+---------+----------+---------+----------+--------+---------------+
| 198         | Connell  | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
| 199         | Grant    | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
| 101         | Kochar   | SH_CLERK | 3400.00 | 1.30     | 118    | 30            |
+-------------+---------+----------+---------+----------+--------+---------------+
```

QUESTION 17: Update job ID to Grade_A for employees earning more than 5000

Update the JOB_ID for employees whose SALARY is greater than 5000. Then display the updated rows.

SQL Query:

```
UPDATE EMPLOYEE
SET JOB_ID = 'Grade_A'
WHERE SALARY > 5000;

SELECT EMPLOYEE_ID, LAST_NAME, JOB_ID
FROM EMPLOYEE
```

```
WHERE SALARY > 5000;
```

Output:

```
Query OK, 4 rows affected

+-------------+-----------+---------+
| EMPLOYEE_ID | LAST_NAME | JOB_ID  |
+-------------+-----------+---------+
| 201         | Hartstein | Grade_A |
| 202         | Fay       | Grade_A |
| 203         | Mavris    | Grade_A |
| 100         | King      | Grade_A |
+-------------+-----------+---------+
```

This concise verification output shows all rows changed by the UPDATE. The full table can also be displayed with SELECT * FROM EMPLOYEE;

QUESTION 18: Display employees who are clerk, programmer or assistant

The job IDs in the supplied table are SH_CLERK (clerk), IT_PROG (programmer) and AD_ASST (assistant). Because Question 17 has already changed the JOB_ID of employees earning more than 5000, the output below reflects the table after that update.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE JOB_ID IN ('SH_CLERK', 'IT_PROG', 'AD_ASST');
```

Output:

```
+-------------+---------+----------+---------+----------+--------+---------------+
 | EMPLOYEE_ID | LAST_NAME| JOB_ID   | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
 +-------------+---------+----------+---------+----------+--------+---------------+
 | 198         | Connell  | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
 | 199         | Grant    | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
 | 200         | Whalen   | AD_ASST  | 4400.00 | 1.30     | 101    | 10            |
 | 206         | Gitz     | IT_PROG  | 5000.00 | NULL     | 103    | 60            |
 | 101         | Kochar   | SH_CLERK | 3400.00 | 1.30     | 118    | 30            |
 +-------------+---------+----------+---------+----------+--------+---------------+
```

5 employees match after Question 17. Hartstein and King no longer have their original job IDs because Question 17 updated them to Grade_A.

QUESTION 19: Display clerks whose salary is less than 3000

SH_CLERK represents clerk in the supplied data. Both conditions must be true.

SQL Query:

```
SELECT *
FROM EMPLOYEE
WHERE JOB_ID = 'SH_CLERK'
AND SALARY < 3000;
```

Output:

```
+-------------+-----------+----------+---------+----------+--------+---------------+
 | EMPLOYEE_ID | LAST_NAME | JOB_ID   | SALARY  | COMM_PCT | MGR_ID | DEPARTMENT_ID |
 +-------------+-----------+----------+---------+----------+--------+---------------+
 | 198         | Connell   | SH_CLERK | 2600.00 | 2.50     | 124    | 50            |
 | 199         | Grant     | SH_CLERK | 2600.00 | 2.20     | 124    | 50            |
 +-------------+-----------+----------+---------+----------+--------+---------------+
```

QUESTION 20: Display last name and manager ID for salary above 3000 under manager 101

Display only LAST_NAME and MGR_ID where SALARY is greater than 3000 and MGR_ID equals 101.

SQL Query:

```
SELECT LAST_NAME, MGR_ID
FROM EMPLOYEE
WHERE SALARY > 3000
AND MGR_ID = 101;
```

Output:

```
+-----------+--------+
 | LAST_NAME | MGR_ID |
 +-----------+--------+
 | Whalen    | 101    |
 | Mavris    | 101    |
 | Baer      | 101    |
```

```
+-----------+--------+
```

END OF DBMS LAB ASSIGNMENT-3